UNITED STATES OF AMERICA
BEFORE THE
FEDERAL ENERGY REGULATORY COMMISSION

Interconnection of Large Loads to the Interstate Transmission System	))	Docket No. RM26-4-000 (Not consolidated)
PJM Interconnection, L.L.C., *et al.*	)	Docket No. EL25-49-000
Large Loads Co-Located at Generating Facilities	)	Docket No. AD24-11-000
Constellation Energy Generation, LLC v. PJM Interconnection, L.L.C.	)))))	Docket No. EL25-20-000 (Consolidated)

SUPPORTING COMMENTS OF CONSTELLATION ENERGY GENERATION, LLC

Constellation welcomes Secretary of Energy Chris Wright's proposed ANOPR[1] and offers these comments to support (with limited qualifications) its fourteen principles "that are intended to ensure efficient, timely, and non-discriminatory [large] load interconnections"[2] and the adoption of a final rule on the Secretary's requested timeline. More immediately, "given the urgency of this issue," the Commission should "build upon these principles"[3] by issuing a full order in the PJM Show Cause proceedings[4] consistent with the ANOPR's guidance based upon the extensive record

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[1] U.S. Department of Energy, Secretary of Energy, Letter to FERC re: "Ensuring the Timely and Orderly Interconnection of Large Loads," Advance Notice of Proposed Rulemaking (Oct. 23, 2025) (issued pursuant to Department of Energy Organization Act § 403, 42 U.S.C. § 7173) (cover letter, "Secretary's Letter," and Advanced Notice of Proposed Rulemaking, "ANOPR").

[2] Secretary's Letter at 1-2.

[3] *Id.* at 2.

[4] *PJM Interconnection, L.L.C.*, Docket No. EL25-49-000, *Large Loads Co-Located at Generating Facilities*, Docket No. AD24-11-000, *Constellation Energy Generation, LLC v. PJM Interconnection, L.L.C.*, Docket No. EL25-20-000 (consolidated) (collectively, "PJM Show Cause proceedings").

already established in that consolidated docket. Alternatively, if the Commission believes it needs additional time to resolve the PJM Show Cause proceedings, it should issue a targeted PJM order that resolves the few most critical issues that continue to delay large load interconnections in PJM.[5] This is the Commission's best path forward to promptly carry out the imperatives of the ANOPR in PJM as the Commission works in parallel on a comprehensive final rule, ensuring that America continues to lead the AI race while preserving reliable service.

EXECUTIVE SUMMARY

Constellation agrees with the motivations and principles outlined in the ANOPR and the need for clear rules to allow the timely interconnection of large loads and their co-location with generators. Constellation urges the Commission to issue a final comprehensive rule, consistent with the ANOPR, by April 30, 2026. We highlight three points at the outset:

First, the Commission should affirm its jurisdiction over large load interconnections to the transmission system generally and to hybrid facilities (*i.e.*, the generator and load sharing an interconnection to the interstate grid) in particular. Federal authority over the interconnection of hybrid facilities is straightforward given that the shared generator interconnection already is Commission-jurisdictional (see ANOPR Principle 1). The Commission should reject any argument by Transmission Owners (or others) that low-voltage facilities that step down the power from the interstate grid at the point of interconnection to the load are outside the Commission's jurisdiction. Contrary to such a contention, the Federal Power Act ("FPA") gives the Commission jurisdiction over "the transmission of electric energy in interstate commerce,"[6] which includes "unbundled

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[5] Today, Constellation is also filing a motion in the PJM Show Cause proceedings to request an immediate Commission order consistent with the ANOPR and these comments.

[6] 16 U.S.C. § 824(b)(1).

retail transmissions."[7] Thus, the Commission has jurisdiction over the timing and procedures applicable to large load interconnections to the transmission system.

Second, the Commission should reject undue discrimination against existing generation. This includes "Bring Your Own Generation" proposals in which only new generation resources are eligible to participate. It also includes reliability studies that would apply disproportionate requirements to existing generation as compared to new generation (see ANOPR Principle 10). Discriminatory policies in favor of new generation inevitably lead to the premature retirements of existing generation, including highly reliable dispatchable and baseload generation, including many nuclear plants that have struggled in recent years to remain economic. Resource adequacy requires both new generation and *retention of existing resources* to ensure the lights stay on and that all load is served. New and existing resources should be able to compete on a level playing field to serve new load.

Third, the Commission should not wait until a final rule to implement the ANOPR principles in PJM, where data center development has been stymied by disagreements and uncertainty over who controls the timing and nature of large load interconnections, and over the terms of any ensuing transmission service. The current load interconnection process in PJM in practice cedes control of all large load interconnections to the discretion of Transmission Owners, regardless of the facilities to which the load proposes to interconnect. Some Transmission Owners have simply refused to participate in PJM's studies to analyze reliability impacts; and some, acting under unfiled interconnection study procedures, have changed their entire construct and timetable

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[7] *New York v. FERC*, 535 U.S. 1, 16 (2002).

for processing load studies.[8] Some Transmission Owners have refused to allow new load connections unless the load agrees to the Transmission Owner's favored configuration or agrees to take other services offered by the Transmission Owner that the load does not need or wish to take and should not be required to take.[9]

The ANOPR proposes to resolve those issues and set forth clear rules for interconnection study processes and for hybrid load and generation configurations connected to transmission facilities—the same issues that were the impetus for Constellation's original complaint filed 12 months ago in Docket No. EL25-20. Constellation supports a final large load interconnection rule on the Secretary's timeline, but in the meantime, the Commission should act first where it already has a fully developed record. In PJM, the Commission still has pending a nine-month-old preliminary finding that "the existing [PJM] Tariff appears to be unjust and unreasonable or unduly discriminatory or preferential because it does not contain provisions addressing with sufficient clarity or consistency the rates, terms, and conditions of service that apply to co-location arrangements."[10] After already soliciting comments in a national proceeding on large loads (Docket No. AD24-11-000) and receiving full briefing on Constellation's complaint seeking clarity in the PJM Tariff (Docket No. EL25-49-000), the Commission further expanded the record in the PJM Show Cause proceeding with 38 comprehensive briefing questions covering issues

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[8] For evidence and discussion of Transmission Owners' refusal to cooperate, *see* Complaint Requesting Fast Track Processing of Constellation Energy Generation, LLC, Docket No. EL25-20-000, at 6-9 (filed Nov. 22, 2024) ("Constellation Complaint") (citing Protest of Constellation Energy Generation, LLC, Docket No. EL24-149-000, Exh. 2 (correspondence with Exelon) (filed Oct. 30, 2024); Motion to Lodge of Constellation Energy Generation, LLC, Docket No. EL24-149-000, Exh. 1 (November 12, 2024 Exelon Letter) (filed Nov. 18, 2024)); *see also* Constellation Complaint, Exh. 5, Declaration of James R. McCoskey (detailing interaction with Exelon to study co-location projects).

[9] *See Atl. City Elec. Co.*, 190 FERC ¶ 61,109 (2025) (rejecting Exelon Transmission Owners' attempt to force all co-located load to take full transmission service).

[10] *PJM Interconnection, L.L.C.*, 190 FERC ¶ 61,115, at PP 74-75 (2025) ("PJM Show Cause Order").

nearly identical to those now raised in the ANOPR.[11] The Commission originally targeted late June 2025 for an order,[12] and has reiterated its intent to resolve that matter expediently.[13] The Commission should now issue a ruling in that pending proceeding implementing the ANOPR's principles, even as it also initiates a rulemaking to apply those principles outside the PJM region.[14]

Certainly, the ANOPR, which is intended to expeditiously resolve some of the most controversial issues confronting large load interconnections, should not be the cause of further delay in PJM. According to the Secretary, the ANOPR "is not intended in any way to discourage public utilities from making filings to address these and similar issues under FPA section 205."[15] If the ANOPR is not intended to discourage public utilities from filing new proposals under section 205, the Commission should not be discouraged from ruling on pending proceedings under section 206 where it already has a comprehensive record, made a preliminary finding that the existing tariff is unjust, unreasonable and unduly discriminatory, and indicated it is poised to act.

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[11] *Compare id.* P 88 (posing 38 briefing questions), *with* ANOPR at ¶¶ 17-32 (positing 14 principles).

[12] PJM Show Cause Order, 190 FERC ¶ 61,115 at P 91 (estimating that the Commission would be able to issue its decision within approximately three months of the filing of Tariff revisions, which the Commission required to be submitted within thirty days of its order (*i.e.*, by March 24, 2025)).

[13] *See, e.g., PJM Interconnection, L.L.C.*, 191 FERC ¶ 61,025, at P 30, n.106 (2025) ("We fully intend to act as expeditiously as possible to address those issues, as we know how important it is to investors, utilities, and all interested parties to have regulatory certainty."); *see* FERC, November 20, 2025 Open Meeting, YOUTUBE (Nov. 20, 2025), at 19:39 https://www.ferc.gov/news-events/events/november-20-2025-open-meeting-10022024 (Commissioner David Rosner expressing he is "confident that we can make some progress" on the open co-location proceeding in PJM); Transcript of Oct. 16, 2025 Open Meeting at 7:14-18 (Then-Chairman David Rosner states "I'm very optimistic that the Commission will be ready to act later this year on our work on colocation. Commissioners See and Chang and I have been working diligently on that, and I'm really happy with the progress we have been making."), available at: https://www.ferc.gov/media/transcript-october-16-2025-open-meeting.

[14] Constellation recognizes that the Commission has undergone changes in leadership and Commissioners in the intervening months, as well as a government shutdown. Now that those transitional issues are behind it, the Commission can focus on completing the proceeding.

[15] ANOPR at ¶ 32.

Accordingly, as the Commission develops a final comprehensive rule on the ANOPR,[16] the Commission should also immediately issue an order in the PJM Show Cause proceedings as set forth below in Section II.A., adopting the Commission's preliminary finding that the PJM Tariff is unjust and unreasonable and establishing a replacement rate based on the record in those proceedings that reflects the ANOPR's guidance, with potential additional measures to enhance resource adequacy and reliability. The Commission should focus on establishing transparent processes that ensure that new large load can be promptly interconnected while reliability is maintained, and on developing transmission rates that (i) are just, reasonable and not unduly discriminatory and (ii) create strong incentives for load to locate in the most efficient places, where network upgrades will be minimized. This approach will serve the dual goals of protecting consumers from unnecessary upgrade costs and timely, effective, and reliable connection of load consistent with the national interest in winning the AI race.

Alternatively, if the Commission chooses not to issue a full replacement rate in the PJM Show Cause proceedings, it should issue a partial replacement rate as set forth below in Section II.B as soon as possible that calls balls and strikes on three critical issues consistent with the record and the ANOPR (all of which should also be part of a comprehensive order on the merits): (1) clarifying jurisdictional boundaries consistent with the ANOPR, (2) rejecting a one-size-fits-all firm transmission regime based on gross rates, and (3) requiring non-discriminatory,

[16] The Commission might also consider whether its rule should focus, at least initially, on the Regional Transmission Organization ("RTO") markets rather than to every Transmission Owner in the nation where, in some cases, the finding under section 206 of the FPA that current rules are unjust and unreasonable may lack sufficient record evidence or present a more difficult showing. In PJM, by contrast, the Commission has a comprehensive record of evidence that the existing PJM Tariff is unjust and unreasonable because it lacks clear rules for co-location and large load interconnections.

timely, and meaningful cooperation by Transmission Owners on processing large load and co-location studies.

COMMENTS

I. Constellation Supports the ANOPR's Fourteen Principles with Limited Qualifications

Constellation supports a final large load interconnection rule arising out of the ANOPR on the Secretary's expedited timeline and offers these comments in support of the ANOPR's fourteen principles, with a few qualifications specified herein.

Principle 1—Confirming Federal Jurisdiction Over Large Load Interconnections to Interstate Transmission Facilities

Constellation agrees with the ANOPR's principle that the Commission has jurisdiction over interconnections directly to transmission facilities.[17] That principle is consistent with the FPA, which gives the Commission exclusive jurisdiction over "the transmission of electric energy in interstate commerce."[18] Interconnection with the interstate transmission grid falls within that scope. As the Commission has long recognized, "interconnection is part and parcel of transmission of electric energy in interstate commerce, and thus interconnection service is part and parcel of jurisdictional transmission service."[19]

Interconnection of hybrid facilities, in particular, provide a crystal clear case for federal authority. The ANOPR defines hybrid facilities as large loads "seeking to share a point of

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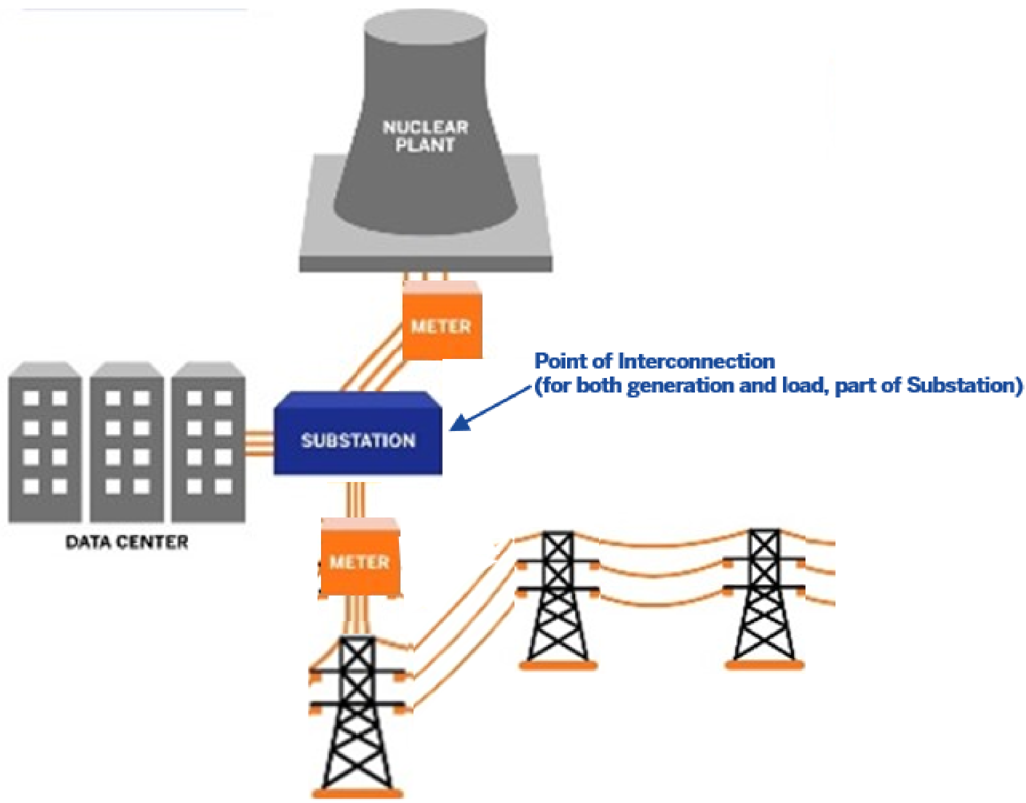
[17] ANOPR at ¶ 18. While the ANOPR adds that this principle is "consistent with the Commission's seven-factor test," the Commission need not resort to that test to conclude that it has jurisdiction over large load interconnections to interstate transmission facilities, as explained further below.

[18] 16 U.S.C § 824(b)(1).

[19] *Pac. Gas & Elec. Co.*, 115 FERC ¶ 61,193, at P 36 (2006); *see also Duke Energy Progress, LLC v. FERC*, 106 F.4th 1145, 114 (D.C. Cir. 2024) ("Among the transactions that FERC regulates is the connection of a new generation facility to the physical infrastructure owned by a Grid Operator."); *Nat'l Ass'n of Regul. Util. Comm'rs v. FERC*, 475 F.3d 1277, 1280 (D.C. Cir. 2007) (holding that, "[b]y establishing standard [interconnection] agreements," FERC properly "exercised its jurisdiction over the *terms* of those relationships") (emphasis in original).

interconnection with new or existing generation facilities."[20] Hybrid large loads are transmission customers that can draw power both from the co-located generator as well as the grid. This hybrid configuration is one that PJM has itself proposed in the PJM Show Cause proceedings (PJM Option 2).[21] It looks like this:



Regulation of the interconnection between a generator and the interstate transmission grid is well within the scope of the Commission's jurisdiction.[22] Because the generator's point of interconnection is FERC-jurisdictional, so too must be the load's. It would be inconsistent for the shared point of interconnection to be FERC-jurisdictional for the generator but not for the large load, and the Commission does not lose its jurisdiction over the interconnection simply because a

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[20] ANOPR at ¶ 12.

[21] *See* April 23, 2025 Constellation Answer at 38.

[22] *Pac. Gas & Elec. Co.*, 115 FERC ¶ 61,193 at P 36.

load also happens to interconnect to the interstate grid at the same point. It is irrelevant that the load is an end-user of electricity. While the retail *sale of energy* directly to a load co-located with a generator at the same point of interconnection is not Commission-jurisdictional, the *transmission* of electricity to the co-located load at the shared point of interconnection is Commission-jurisdictional.

Indeed, the Supreme Court has long made clear that the Commission has jurisdiction over *all* transmissions, including transmissions to end users. "[T]he FPA authorizes FERC's jurisdiction over interstate transmissions, without regard to whether the transmissions are sold to a reseller or directly to a consumer."[23] That is because "[t]here is no language in the statute limiting FERC's *transmission* jurisdiction to the wholesale market, although the statute does limit FERC's *sale* jurisdiction to that at wholesale."[24] Accordingly, as the Commission has itself recognized, it has jurisdiction over interconnections "based on the nature of interconnection to the transmission system generally, not the purpose for such interconnection (*i.e.*, to make a wholesale sale or to make a wholesale or retail purchase)."[25] Therefore, the transmission of energy from the interstate grid to a hybrid facility, and the interconnection of that facility to the grid, is well within the Commission's jurisdiction. It does not matter that the energy is ultimately transmitted to an end user (*i.e.*, the large load) because the Commission's *transmission* jurisdiction is not "specifically confined to the wholesale market."[26]

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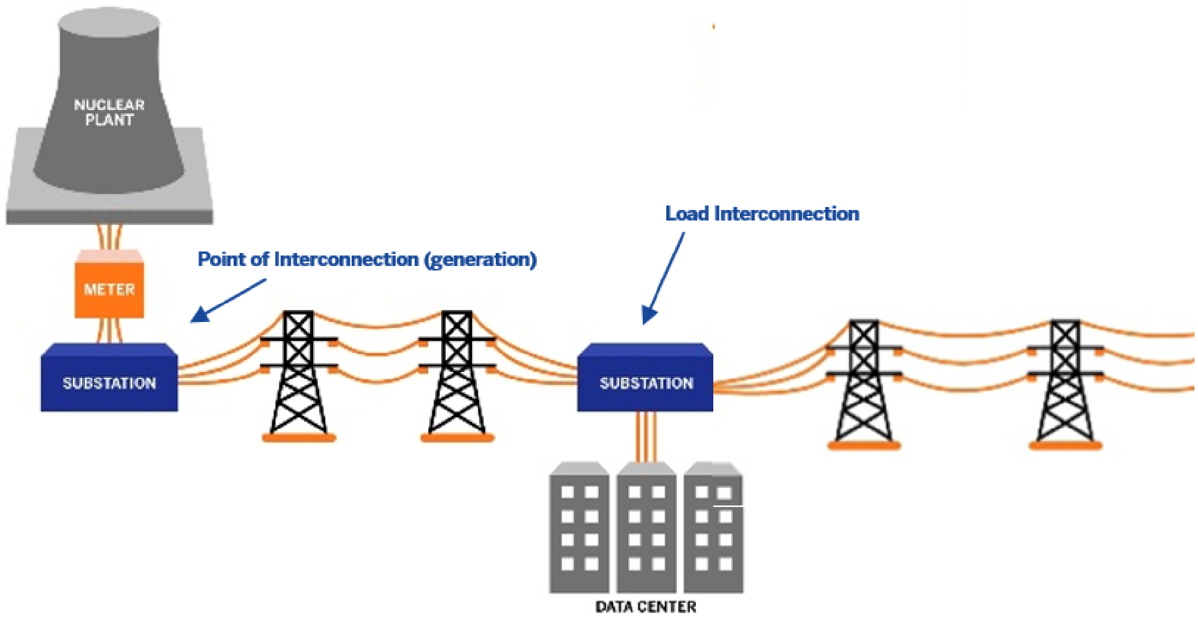
[23] *New York v. FERC*, 535 U.S. at 20.

[24] *Id.* at 17.

[25] PJM Show Cause Order, 190 FERC ¶ 61,115 at P 72.

[26] *Id.* at 20; *see also Nat'l Ass'n of Regul. Util. Comm'rs*, 475 F.3d at 1282 (upholding Commission's jurisdiction over a "transmission facility [that] also engages in local distribution … insofar as the interconnections are 'for the purpose of making sales of electric energy for resale in interstate commerce" (quoting *Standardization of Generator Interconnection Agreements & Procedures*, Order No. 2003, 104 FERC ¶ 61,103, at P 804 (2003) ("Order No. 2003"))).

While hybrid facilities present a straightforward case for Commission jurisdiction, the Commission also has jurisdiction over the interconnection of large loads to transmission facilities even when they are not part of a hybrid facility or co-located with a generator, as in this image:



The ANOPR presents a compelling case for federal jurisdiction over the interconnection between a large load and the interstate transmission grid, and its reasoning does not depend on whether the large load is a hybrid facility.[27] The nature of the *interconnection* between a large load and the interstate grid is no different from the nature of the interconnection between a hybrid facility and the interstate grid. Each is "part and parcel of transmission of electric energy in interstate commerce,"[28] and the Commission is fully within its jurisdiction to regulate it.

The Commission should reject any argument that low-voltage facilities that step down the power from the interstate grid at the point of interconnection to the load are outside the Commission's jurisdiction. Such step-down facilities are necessary components of the

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[27] ANOPR at ¶¶ 13-16.

[28] *Pac. Gas & Elec. Co.*, 115 FERC ¶ 61,193 at P 36.

interconnection—a clear transmission facility. The FPA's plain text gives the Commission jurisdiction over "the transmission of electric energy in interstate commerce,"[29] including transmissions to end-users, and to remove the step-down facilities from the Commission's jurisdiction would improperly limit the full scope of transmission jurisdiction that the FPA has conferred on the Commission.[30] Moreover, as a practical matter, all large loads require step-down facilities to interconnect with the grid, so to hold that such step-down facilities are beyond the Commission's jurisdiction would effectively allow Transmission Owners to retain their control over the timing, procedures and services applicable to large load interconnections, thereby thwarting the entire purpose of the ANOPR. That cannot be squared with the FPA's conferral of jurisdiction to the Commission over the transmission of electricity, including to end-users, as the Supreme Court has held. The Commission should issue clear guidance on this point, lest the Transmission Owners use litigation over the seven-factor test to thwart the intent of the ANOPR.

At the same time, the ANOPR correctly limited the Commission's jurisdiction to "interconnections directly to transmission facilities," not retail sales.[31] Under the FPA, only the states, not the Commission, may exercise jurisdiction over the sale of energy to end-users.[32] The Commission therefore has no jurisdiction over the direct sale of energy from a generator to a data center where the power never flows over the interstate transmission grid. This includes, for example, a behind-the-meter co-location configuration with system protection features that prevent any power from flowing over the grid to the load, like this:

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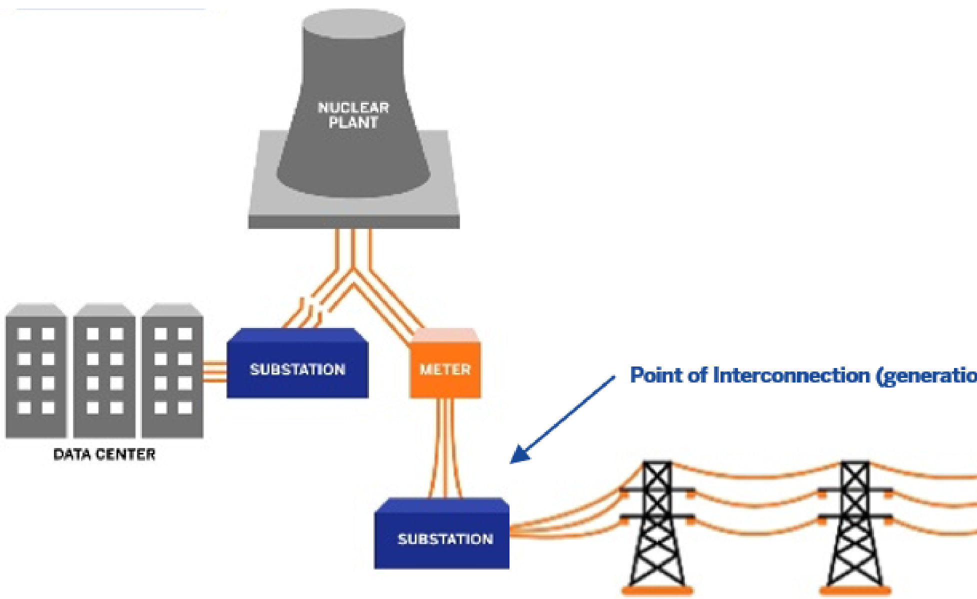
[29] 16 U.S.C. § 824(b)(1).

[30] *New York v. FERC*, 535 U.S. at 16.

[31] ANOPR at ¶ 18.

[32] *See* 16 U.S.C § 824(b)(1).



As the Commission correctly observed—in line with the ANOPR—"states retain exclusive jurisdiction over the terms of retail sales, generally including the rate designs that determine how the costs of the wholesale sale and transmission of electricity assigned to a wholesale customer are allocated among that wholesale customer's retail customers."[33] This is equally true for the sale between the generator and a load connected behind the generator meter in a configuration that makes it impossible for the load to draw power from the grid.

The Commission reiterated this principle just this past month in *Tri-State Generation and Transmission Association, Inc.*[34] There, the Commission rejected Tri-State's proposed tariff for large loads because it unlawfully "intru[ded] on retail rate regulation" and thus was outside "the Commission's authority."[35] As the Commission stated, "the plain language of the FPA bars the Commission from regulating retail sales and the Supreme Court has been clear: specification of

[33] *Pac. Gas & Elec. Co.*, 115 FERC ¶ 61,193 at P 68.

[34] 193 FERC ¶ 61,070 (2025) ("*Tri-State*").

[35] *Id.* P 45.

terms of sale at retail 'is a job for the States alone.'"[36] That encompasses direct sales from generators to large loads.[37]

Principle 2—Setting an Appropriate Size Threshold

The ANOPR proposes a 20 MW size threshold, or "for hybrid facilities, where the load is greater than 20 MW.[38] Constellation supports the ANOPR's size threshold concept, but it is unnecessary for hybrid facilities where the load shares an interconnection to the transmission grid with generation as all hybrid facilities' load is FERC-jurisdictional. For large loads not sharing an interconnection with generation, it may be that the size threshold should be based on the voltage of the transmission facilities where it connects. But a higher megawatt (or voltage) threshold would have the significant benefit of reducing the potential number of large load projects clogging the interconnection queue.

Principle 3—Combining Studies for Load and Hybrid Facilities with Generation

The ANOPR states that "to the extent practicable, load and hybrid facilities should be studied together with generating facilities. Such an approach will allow for efficient siting of loads and generating facilities and thereby minimize the need for costly network upgrades."[39] It is our understanding that a single, combined study is *not* practicable, as the generation and large loads typically still must be studied separately in case the load or generator trips, and either the generator

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[36] *Id.* P 50 (citing *FERC v. Elec. Power Supply Ass'n*, 577 U.S. 260, 280 (2016); *New York v. FERC*, 535 U.S. at 17).

[37] In *Tri-State*, the Commission had no occasion to "elaborate … on other matters raised by parties regarding the extent of the Commission's jurisdiction, such as jurisdiction over the process of interconnecting loads to the Commission-jurisdictional transmission system." *Id.* As explained above, the Commission's jurisdiction does in fact extend to interconnections between large loads and interstate transmission facilities.

[38] ANOPR at ¶ 19.

[39] *Id.* at ¶ 20.

injects its entire output onto the grid, or the load withdraws its full demand from the grid.[40] The Commission should follow the guidance of the RTOs regarding the technicalities of the studies, including whether they can be combined for hybrid facilities with a shared point of interconnection to the grid.

The "benefits" of combined studies are not as certain as they may appear: *First*, the goals of efficient siting of load and minimizing upgrades should be priorities, but these are benefits of *co-location*, not combining studies. Indeed, other measures would have a greater impact on efficient siting and minimizing upgrades than combining studies, including:

- confirming hybrid facilities (generation and load sharing an interconnection to the grid) should pay transmission rates on a net load basis (or standby charges, both as discussed below at Principle 11),
- approving large load interconnection standards (see Principle 4), and
- adopting and enforcing strict study timelines, non-discrimination, and cooperation (see Section II.B. below).

Second, combined studies are frequently paired with "bring your own generation" (BYOG) proposals that are restricted to new generation and unduly discriminate against existing generation resources. The ANOPR itself suggests this, stating that "siting a large load near or at the same point of interconnection as a *new* generating facility could reduce the network upgrades needed to interconnect only the load or only the generating facility."[41] But all co-location configurations

[40] A combined study is appropriate for a behind-the-meter configuration where the large load connects behind the generator's meter (in contrast to hybrid facilities where the load and the generator share a point of interconnection to the grid "in-front-of-the meter"). In a behind-the-meter configuration, the generator modifies its Commission-jurisdictional interconnection agreement and the studies are based on the net rights to the transmission system.

[41] ANOPR at ¶ 20 (emphasis added); *see also id.* at ¶ 22 (studying hybrid facilities' injection and/or withdrawal rights "provides incentives for co-location with *new* generation facilities and ensures efficient buildout of the transmission system") (emphasis added); *id.* at ¶ 27 (Principle 10 calling for a unique study methodology to apply to "an *existing* generation facility in a hybrid facility arrangement) (emphasis added); *but see id.* at P 12 (defining "hybrid facilities" as large loads "seeking to share a point of interconnection with *new or existing* generation facilities") (emphasis added). There should be no discrimination between new and existing generation in the co-location rules that the Commission adopts.

provide these benefits, not just those linked to new generation. The Commission should reject proposals that would allow a separate load study or load interconnection process solely for load coupled with new resources or any option that would turn combined studies into a vehicle for compelled BYOG arrangements limited to new resources only.

This issue was raised in the PJM Show Cause proceedings, which included a potential BYOG proposal (PJM Option 6) focused on PJM extending favorable *generation* interconnection queue treatment to new generators associated with new loads.[42] Constellation provided testimony that "the suggestion that new generation that directly contracts with a customer … should be allowed to skip the generation interconnection queue undermines PJM market signals.… These generation developers will be, in effect, allowed to circumvent the queue process … in a manner that is unfair and not competitive."[43] "This approach is functionally equivalent to an out-of-market payment or subsidy (for the same resource adequacy services) which effectively lowers the cost of the with-contract class and thus distorts and depresses market economics for all other resources (including all without-contract new resources and all existing resources)."[44] The favorable interconnection queue treatment presented by PJM Option 6 does not make up for the discriminatory nature of this option because "[t]he ultimate endpoint of such a policy is to drive all resource additions to seek contracts with large load and depress economics for other resources, leading to reduced new entry outside of the large load contract path and potentially retirements of existing resources."[45]

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[42] Constellation's April 23, 2025 Show Cause Answer at 52–55 (Option 6—"Bring Your Own Generation").

[43] Testimony of Zachary Ming at 58-59 (Exh. 2 to Constellation's April 23, 2025 Show Cause Answer) ("Ming Testimony").

[44] Affidavit of Aaron T. Patterson at ¶ 40 (Exh. 3 to Constellation's April 23, 2025 Show Cause Answer) ("Patterson Affidavit").

[45] Constellation's April 23, 2025 Show Cause Answer at 52-53 (quoting Patterson Affidavit at ¶ 40).

However, as bad as PJM's Option 6 is, far worse would be a rule that provides expedited *load* interconnection treatment to loads that bring their own new generation, effectively creating a powerful disincentive for new load to contract with existing resources. This would be unlawful discrimination and bad policy,[46] yet that is what has been discussed in the PJM Stakeholder Process. It takes more than just new resources to supply the power system—we also must retain existing resources and not create policies that cause their premature retirement, such as distorting market signals by paying above-market premiums or offering special favors solely to new resources. In a fight for AI dominance with China, the United States cannot afford to tie one hand behind its back.[47]

The simple solution is to allow existing resources to compete on an equal footing with new resources and to ensure that the final rule treats existing and new resources without discrimination. The Commission should affirm that all market design features should be designed to ensure sufficient generation resources, which means to provide efficient signals both to build new capacity *and* to retain existing resources. BYOG fails this basic test, and the Commission should reject a "combined studies" approach as part of a BYOG-like mechanism. In PJM, the Commission

[46] *See* Constellation's April 23, 2025 Show Cause Answer at 53-55 (discussing unlawful "vintaging" in favor of new resources).

[47] In the PJM Show Cause proceedings, seven Retired Generals of the United States Armed Services, stated, "nuclear plants serving co-located data centers provide a reliable source of power and are uniquely hardened against cyber and physical attacks…. Co-location arrangements provide crucial advantages in terms of the speed with which they can meet defense-critical data center needs, and the resilience of the power that nuclear plants provide. We respectfully request that you take these benefits into account as FERC's proceedings go forward and ensure that all options to quickly serve data centers are made available." National Security Experts Letter to FERC, Docket No. EL25-49-000, at 2 (Apr. 22, 2025) (Accession No. 20250422-5129). *See also* Declaration of Dr. Paul Stockton at 3, 9 (Exh. 1 to Constellation's April 23, 2025 Show Cause Answer) ("Stockton Declaration") ("[m]eeting the power requirements for these data centers is crucial so that [the Department of War] can … counter China's drive for AI supremacy and its use of AI to help achieve military superiority over the United States.… Speed is of the essence for national security.").

should reject PJM Option 6 or any option that provides expedited load interconnection in exchange for BYOG as a potential pathway.

Principle 4—Adopting Standardized Interconnection Rules and Expediting Service to Large Loads

The ANOPR's fourth principle proposes standardized study deposits, readiness requirements, and withdrawal penalties to large loads and hybrid facilities.[48] While the details matter, this general approach mirrors the proven framework established in Order No. 2003, which standardized generator interconnection procedures by requiring deposits, readiness demonstrations, and withdrawal penalties.[49] Extending the same principles to large loads and hybrid facilities should improve queue efficiency, enhance forecasting accuracy, and support more effective transmission planning.

The lack of consistent standards for large loads and hybrid facilities creates a vacuum that leaves Transmission Owners with significant discretion to delay or deny large-load and co-location proposals and to impose terms that are neither just nor reasonable.[50] The same types of Transmission Owner discrimination and delays are what caused the Commission to adopt Order No. 2003 for generator interconnections.[51] These are the very concerns surfacing today in large-

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[48] ANOPR at ¶ 21.

[49] Order No. 2003, 104 FERC ¶ 61,103.

[50] *See, e.g.*, Constellation's April 23, 2025 Show Cause Answer at 14–15 (discussing Transmission Owner refusals to process Necessary Studies and requirements for the projects to be in front of the meter and to take service on a gross load basis).

[51] *See, e.g.*, Order No. 2003, 104 FERC ¶ 61,103 at P 120 (a Transmission Provider "may not require an Interconnection Customer, as a condition of interconnection, to accept responsibility for Network Upgrades on other systems," because doing so "would create an unreasonable obstacle to the construction of new generation"); *id.* P 115 (requiring transparency on slippage by directing Transmission Providers to "post deviations from the study time lines … along with an explanation for the delay"); *id.* P 353 (adopting an Option to Build to mitigate utility-caused schedule risk "the Interconnection Customer may assume responsibility for the construction of the Transmission Provider's Interconnection Facilities and Stand Alone Network Upgrades if the Transmission Provider notifies [it] that it cannot meet the dates established") (we discuss the Option to Build in Principle 9).

load interconnections: absent clear, filed standards, Transmission Owners are again blurring interconnection with other services, imposing delay leverage, and conditioning progress on unrelated obligations—conduct Order No. 2003's reforms were designed to prevent. The Commission should direct Transmission Owners to process transmission interconnection requests in a non-discriminatory fashion, regardless of the configuration (*i.e.*, a hybrid front-of-the meter connection or a direct connection to the grid, taking service from the Transmission Owner).

Another current example of Transmission Owners leveraging their control over interconnection is in the negotiation of Transmission Security Agreements ("TSAs"). While Constellation is supportive of the concept of TSAs to deter speculative load interconnection requests, it is important that those TSAs contain some component reflective of upgrade costs (to create an incentive to locate most efficiently), and that they do not discourage legitimate projects by locking in one-size fits all gross billing arrangements.

The ANOPR's Principle 4 further highlights the benefits of "provid[ing] transmission providers with more useful information to more accurately forecast demand on their systems."[52] PJM, for example, currently relies on Transmission Owners' annual assessments, which are informal and based on unverified estimates of anticipated additions. In the PJM Show Cause proceedings, Constellation advocated mandatory, detailed rules for Transmission Owner reporting of anticipated load additions, with reporting at least two to four times per year to keep planning assumptions.[53] Building on that foundation, Constellation, other generators, and several large technology companies with large load needs have sponsored a "Joint Stakeholder Package,"

[52] ANOPR at ¶ 21.

[53] Constellation's April 23, 2025 Show Cause Answer at 96-97; *see also* Patterson Affidavit at ¶ 29

discussed in Principle 14 below on reliability standards, which includes a large load forecasting proposal to ensure only substantiated load is reflected in planning baselines.[54]

Given the acute and demonstrable problems with large load forecasting within PJM, the Commission should adopt clear, uniform standards, which should also consider large load forecasting standards and other measures that will expedite large load studies and interconnections. In the PJM Show Cause proceedings, the Commission could order PJM and Transmission Owners to adopt large load forecasting improvements along the lines of the Joint Stakeholder Package's Large Load Forecasting Improvements proposal.

Principle 5—Basing Interconnection Studies for Hybrid Facilities on Withdrawal and/or Injection Rights

Constellation agrees in principle with concept of studying hybrid facilities and any large load configuration "based on the amount of injection and/or withdrawal rights requested,"[55] or reflecting the net usage of the transmission system, but will defer to the RTOs as to what studies are necessary to maintain reliability. Our understanding is that exactly how the studies should work depends on the type of interconnection configuration.

For hybrid facilities, which the ANOPR defines as having a shared point of interconnection "directly to transmission facilities"—that is, an in-front-of-the-meter connection to Commission-jurisdictional interstate transmission facilities—it is our understanding that, as a practical matter, the large load's withdrawals *and* the generator's injections typically still must be studied separately in case the load or generator trips (see discussion above in Principle 3). A load connected in front

[54] *See* Joint Stakeholder Package (Amazon, Calpine, Constellation, Google, Microsoft, Talen), PJM CIFP – Large Load Additions, Stage 4 (Nov. 19, 2025), Exh. 1 hereto (describing "Large Load Forecasting Improvements" proposal); *see also infra* at Principle 14 (discussing same).

[55] ANOPR at ¶ 22.

of the meter seeks withdrawal rights equal to its gross load, and the generator requires injection rights equal to its gross capacity. That said, transmission *rates* for hybrid facilities should be based on the actual *usage* of the transmission system, rather than the load's withdrawal *rights*. And a hybrid facility's actual usage of the transmission system is best measured by net withdrawals.[56]

By contrast, for a behind-the-meter co-location configuration in which the generator is connected to the grid but the load is *not*, both the generation and the load can and should be studied together based on their net *rights* (injections and withdrawals) to use the transmission system.[57]

A curtailable load, meanwhile, could have withdrawal rights of zero if it is fully curtailable, even if it is a hybrid facility *(i.e.,* sharing a point of interconnection with the generator). In that scenario, there would be no firm load to be served and, therefore, no network upgrades required until the transmission service converts to a firm service. (We discuss curtailable loads and dispatchable generators associated with hybrid facilities below in Principle 7.)

Contrary to the views of some Transmission Owners, the interconnection studies—and the services offered—should depend on the specific configuration at issue, rather than an assumption that the load must always have full withdrawal rights. This is consistent with how the Commission treats the interconnection of storage facilities, which requires Transmission Owners to conduct studies based on the customer's proposed operating assumptions whether part of a stand-alone, co-

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[56] "[T]he starting point for a co-located load configuration should be a net billing construct that measures the actual energy flowing between the entire co-location configuration and the transmission system." Constellation's April 23, 2025 Show Cause Answer at 39.

[57] In the PJM Show Cause proceedings, Constellation explained that "[f]or generators seeking to serve co-located load under PJM Options 4 and 5 [behind-the-meter, generator-contingent options], the generators' [Interconnection Service Agreement] would be modified … and PJM would perform a [N]ecessary [S]tudy … with the costs of any… upgrades … typically being borne by the co-locating parties." *See* Constellation's April 23, 2025 Show Cause Answer at 94.

located or hybrid configuration.[58] Just as the Commission has required for storage resources, the final large load interconnection rule should reflect the withdrawal rights and different types of service that large loads and hybrid facilities wish to take, rather than requiring every resource to take full network service whether they need it or not. In the PJM Show Cause proceedings, this would be reflected in a Commission order upholding behind-the-meter, generator-contingent services (PJM Options 4 and 5) and curtailable service (PJM Option 7), in addition to Firm Network Integrated Transmission Service ("NITS") service (PJM Option 2).

Principle 6—Installing Appropriate System Protection Facilities

Constellation agrees that a behind-the-meter facility that does not intend to take power from the grid "sh[ould] be required to install the system protection facilities necessary to prevent unauthorized injections or withdrawals that exceed the respective rights."[59] There are other circumstances in which system protection schemes are unnecessary, such as when the load that is part of a hybrid configuration intends to take network integration transmission service. Operational limitations associated with the particular configuration would be set forth in the agreement(s) governing the interconnection, again just like the Commission has required for storage resources.

In PJM, some Transmission Owners have contended that the use of protection schemes for co-located load configurations is inadequately specified and unreliable, citing "the recent history of faults at the Susquehanna generator that has led to unintended power withdrawals from the PJM

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[58] *Improvements to Generator Interconnection Procedures & Agreements*, Order No. 2023, 184 FERC ¶ 61,054, at P 1509 (2023), *order on reh'g & clarification*, Order No. 2023-A, 185 FERC ¶ 61,117 (2023), *order on reh'g & clarification*, Order No. 2023-B, 186 FERC ¶ 61,204 (2024); *see also id.* P 1525 ("[W]e … clarify that the transmission provider must not assign network upgrade costs to the interconnection customer based on those worst-case operating assumptions (e.g., charging at maximum capacity during peak load conditions) where there is agreement from the interconnection customer to, if required, implement operating restrictions including installing or demonstrating that the generating facility already has control technologies … to limit its operations during peak load conditions.").

[59] ANOPR at ¶ 23.

transmission system."[60] This assertion overlooks the corrective and operational safeguards already adopted by PJM and the generating facility. A single historical fault that produced an unintended withdrawal does not demonstrate that protective relays cannot prevent such events. The record in the PJM Show Cause proceedings demonstrates that "[p]rotection schemes are regularly employed on the PJM transmission system, and the successful operation of such protective systems is a common observance within PJM."[61] These systems operate effectively to isolate behind-the-meter co-located loads in the event of a sudden generator outage, ensuring that "no energy will flow to the co-located load over the transmission grid following a contingency, and no NITS service will be provided by the grid to the co-located load."[62] To address the risk of any misplacement or faulty relay systems, such as the Susquehanna situation, the protective scheme should be tested upon installation, similar to other electrical equipment.

There is also evidence in the record in the PJM Show Cause proceedings on "the minimum technical requirements for such system protection facilities."[63] "These protective relay schemes can be designed so that there are completely redundant primary and backup protective schemes, with two circuit breakers placed in series. Each circuit breaker has two trip coils, and all elements are redundant, resulting in a single-failure-proof design."[64]

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[60] *Protest of Exelon Corporation and American Electric Power Service Corporation re the Non-conforming Interconnection Service*, Docket No. ER24-2172, at 12 (June 24, 2024).

[61] Herling Affidavit at ¶ 23 (Exh. 4 to Constellation's April 23, 2025 Show Cause Answer) ("Herling Affidavit").

[62] *Id.* ¶ 29; *see also supra* note 57 (citing Order No. 2023 where the Commission approved control schemes to prevent batteries from drawing power times of grid stress).

[63] ANOPR at ¶ 23.

[64] *Herling Affidavit* ¶¶ 21–22.

Principle 7—Expediting Interconnection Studies for Curtailable Loads

The ANOPR proposes that interconnection studies for "large loads that agree to be curtailable and hybrid facilities that agree to be curtailable and dispatchable should be expedited."[65] Constellation agrees with this principle, but there are additional ways to advance curtailable and dispatchable large loads and hybrid facilities along with expediting their interconnection studies that the Commission should seek to encourage. Curtailability and dispatchability of large loads help maximize efficient use of the transmission system, avoid or defer system transmission (and potentially distribution) upgrades, and reduce capacity procurements.[66]

The recent study by the Nicholas Institute at Duke University explained the benefits of flexible loads and permitting them to take non-firm transmission service. Firm service assumes full transmission service all the time, which curtailable loads *do not need* and *should not be required to buy*—yet it currently is the only network transmission service available. The Duke Study explains that, if new load is required to take firm service and one assumes that such load will "operate at 100% of its maximum electricity draw at all times, including during system-wide peaks"—which is the assumption underlying traditional NITS—then it is "far more likely" that "significant upgrades, such as new transformers, transmission line reconductoring, circuit breakers, or other substation equipment" will be required.[67] By contrast, if a different service were

[65] ANOPR at ¶ 24.

[66] Sam Walsh, *et al.*, *How DOE's Proposed Large Load Interconnection Process Could Unlock the Benefits of Load Flexibility: Considerations for Rulemaking* at 9 (Nov. 2025), https://nicholasinstitute.duke.edu/sites/default/files/publications/how-does-proposed-large-load-interconnection-process-could-unlock-benefits-load-flexibility.pdf ("Duke Policy Statement").

[67] Tyler H. Norris, *et al.*, *Rethinking Load Growth: Assessing the Potential for Integration of Large Flexible Loads in US Power Systems* at 5 (Feb. 2025), https://nicholasinstitute.duke.edu/publications/rethinking-load-growth ("Duke Study").

offered to such new load by which the load could "temporarily reduce (*i.e.*, curtail) its electricity consumption from the grid during [] peak stress periods … it may be able to connect while deferring—or even avoiding—the need for certain upgrades."[68] The Duke Study concludes that a new load curtailment rate of just 0.5% annually would result in nearly 18 gigawatts ("GW") of potential new load integration in PJM without capacity expansion.[69] This amount increases to 23.6 GW with a 1.0% curtailment rate. Finally, the Duke Study also explains that a failure to use existing headroom on the transmission system, reflecting the gap between the average and peak demand in the balancing area, increases costs on ratepayers. Specifically, a low system utilization rate (*i.e.*, more headroom) means that ratepayers are paying to build and maintain generation assets, transmission infrastructure, and distribution networks that may remain idle for much of the year.[70]

While expediting the interconnection studies of flexible load is a good step, the Commission should endorse other paths that would allow curtailable large loads and dispatchable hybrid resources to connect without first requiring the completion of system upgrades needed to support firm transmission service at peak, and without requiring any additional capacity.[71] For example, if the customer chooses, interruptible service can serve as a bridge to full service while transmission upgrades are completed.[72] Importantly, the type of interruptible transmission service discussed here is different from having the interruptible load count as demand response in a

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[68] *Id.* (citing ERCOT, *Large Loads—Impact on Grid Reliability and Overview of Revision Request Package*, Presented at the NPRR1191 and Related Revision Requests Workshop (Aug. 16, 2023), https://www.ercot.com/files/docs/2023/11/08/PUBLIC-Overview-of-Large-Load-Revision-Requests-for-8-16-23-Workshop.pptx).

[69] *Id.* at 2.

[70] *Id.* at 6.

[71] *See, e.g.*, Duke Policy Statement at 12-15 (discussing non-firm withdrawal rights).

[72] *See* ANOPR at ¶ 22, n.39 ("Any facility should be able to seek additional rights through a separate interconnection request.").

capacity auction. It is also different from PJM's ill-fated non-capacity backed load ("NCBL") proposal, as rather than PJM choosing to curtail select classes of customers in a discriminatory manner inconsistent with sections 205 and 202(g) of the FPA,[73] in this instance the load is electing to receive a lower quality transmission service.[74]

In the PJM Show Cause proceedings, PJM identified a provisional, interruptible transmission service, PJM's Option 7, that can be curtailed by PJM during pre-defined conditions and that would result in a more efficient use of the system (*i.e.*, increased system utilization rate) and significant near-term customer savings as compared to the NITS-only framework.[75] Under this option, the interconnection study should be expedited because an "interruptible co-located load does not require the transmission system to be planned, built, or adapted in any way to accommodate the load because PJM can interrupt the load when needed to avoid system impacts. Therefore, no standby or system impact charges are appropriate."[76] "PJM's right to curtail the load during defined criteria also means that the load, while it is receiving this service, need not be counted toward the region's reliability requirement."[77] Large loads also should be able to transition from curtailable service to firm service with appropriate studies and network upgrades, with lesser levels of service allowing more immediate interconnection. Other related market design features

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[73] 16 U.S.C. §§ 824a(g), 824d.

[74] *See* PJM, Large Load Additions: PJM Conceptual Proposal and Request for Member Feedback (Aug. 18, 2025), at https://www.pjm.com/-/media/DotCom/committees-groups/cifp-lla/2025/20250818/20250818-item-03---pjm-conceptual-proposal-and-request-for-member-feedback---presentation.pdf (proposing non-capacity backed load service); *see also* Devin Leith-Yessian, *PJM Drops Non-capacity Backed Load, Shifts Focus to Resource Queue, PRD*, RTO Insider (Oct. 9, 2025), PJM Drops Non-capacity Backed Load, Shifts Focus to Resource Queue, PRD.

[75] Constellation's April 23, 2025 Show Cause Answer at 55-57. Constellation proposed certain modifications, including that interruptible service would be available until the transmission upgrades needed for NITS service are completed.

[76] Ming Testimony at 34.

[77] Constellation's April 23, 2025 Show Cause Answer at 9; *see also id.* at 55-56

can further maximize current use of the system. In the PJM stakeholder process, Constellation has joined a group that proposed the Joint Stakeholder Package of reforms for this purpose, including improving system reliability.[78] The Commission's rule should also consider such measures.

Finally, the ANOPR inquires whether serial (first-come, first-served) studies are appropriate for curtailable loads, and whether a 60-day deadline is reasonable. In Constellation's view, a 60-day period for studying facilities that are likely, by design, to have little impact on the PJM system is aligned with the overall timeline of 6 months for the study of a load interconnection project.

The Commission's proposed rule should adopt these measures to maximize efficient use of the transmission system. In the PJM Show Cause proceedings, the Commission should approve PJM Option 7 as modified by Constellation.

Principle 8—Allocating Network Upgrade Costs Based on Appropriate Studies

The ANOPR's eighth principle states that "load and hybrid facilities should be responsible for 100% of the network upgrades that they are assigned through the interconnection studies" and "seek[s] comment on whether such costs should be offset through a crediting mechanism."[79] Constellation takes no position on this issue at this time.

Principle 9—Granting the Customer the Option to Build

The ANOPR's ninth principle affords large load interconnection customers an option to build equivalent to what is currently provided to generator interconnection customers.[80]

[78] *See* Exh. 1; *see also infra* at Principle 14 (discussing same).

[79] ANOPR at ¶ 25.

[80] ANOPR at ¶ 26; *see Reform of Generator Interconnection Procedures and Agreements*, Order No. 845, 163 FERC ¶ 61,043, at PP 85-87 (2018) ("Order No. 845"), *order on reh'g*, Order No. 845-A, 166 FERC ¶ 61,137 (2019), *order on reh'g & clarification*, Order No. 845-B, 168 FERC ¶ 61,092 (2019).

Constellation agrees that the option to build should be provided to large load customers subject to host utility standards as it is currently provided to generator interconnection customers. This is consistent with the Commission's intentions as stated in its generator interconnection rule, Order No. 845, "of providing interconnection customers more control and certainty during the design and construction phases of the interconnection process."[81] Originally, the option to build was limited "to circumstances where the transmission provider cannot meet the interconnection customer's requested dates" but the Commission found this *too* limiting since the customer may be able to build the facilities not only more quickly but also at lower cost.[82] The Commission found "that in circumstances where an interconnection customer cannot exercise the option to build, it may pay more and/or wait longer for the construction of the transmission provider's interconnection facilities," citing record evidence that "savings can occur and have already occurred."[83] The option to build has been a success and it should be included in the final large load interconnection rule.

Principle 10—Assigning Appropriate Upgrade Costs to Co-Locating Existing Generation

The ANOPR states that "an existing generating facility that seeks to enter a partial suspension to serve a new load at the same location must go through a system support resource (SSR)/reliability must run (RMR) type study," wait to proceed until "after any network upgrades needed to ensure reliability are placed into service," and pay for any upgrades.[84] By "partial

[81] Order No. 845, 163 FERC ¶ 61,043 at P 85.

[82] *Id.* ("The limitation restricts an interconnection customer's ability to efficiently build the transmission provider's interconnection facilities and stand alone network upgrades *in a cost-effective manner*, which could result in higher costs for interconnection customers.") (emphasis added).

[83] *Id.* P 86 (citing a NextEra affiliate's ability to complete a project "one year sooner and for $6 million less than estimated by the transmission provider").

[84] ANOPR at ¶ 27.

suspension," Constellation assumes that the ANOPR is referring to situations in which the RTO requires the generator to delist capacity when a generator serves load directly behind-the-meter and the load takes no transmission service. In PJM, there is an existing "reliability type study" called the Necessary Study process that applies to a circumstance where the generator is directly serving the load. The Necessary Study determines the reliability impact of an existing generator modifying its interconnection, including by co-locating with a large load, with the generator ultimately being responsible to pay for any reliability upgrades necessary to maintain the generator's electrical support of the grid.[85] Constellation supports this existing Necessary Study process and agrees with the principle identified in the ANOPR that the generator is responsible for the upgrades the Necessary Study determines are necessary to maintain operational security and reliability.

To the extent the ANOPR suggests that an existing generator must undergo an SSR/RMR type study if it is required to delist before serving non-network load co-located behind the meter, that should be rejected. It would depart from RTO practice, misapply the existing "retirement" paradigm, and would be unduly discriminatory against existing resources seeking to serve co-located non-network load. The Commission has no authority to force a capacity resource that is required to delist in order to directly serve a non-network customer to instead serve network load.

First, the record in the PJM Show Cause proceedings demonstrates that PJM's existing Necessary Study process is sufficient to ensure reliability for a co-location configuration at an existing generation facility. Constellation's witness, Michael Kormos, PJM's former Chief Operations Officer and former Senior Vice President of Transmission and Compliance at Exelon

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[85] Constellation's April 23, 2025 Show Cause Answer at 95; *see also* PJM's March 24, 2025 Answer at 34.

with nearly 30 years' experience operating and planning transmission systems and running electricity markets, "disagree[d] with any suggestion that [PJM's] [N]ecessary [S]tudy process is anything but a robust, highly effective study of the impact of the modification requested," testifying that the studies are "led by an independent RTO; uniform and transparent … and focused on the overall grid impacts of the new load explicitly (and not focused predominantly on local impacts…)."[86] There is no evidence that this existing framework has failed to maintain reliability in PJM.[87]

Second, treating an existing unit that continues to run and provide electricity to a co-located load as if it were *retiring* is unprecedented and factually incorrect. Instead of retiring, these units continue to serve load. There is no contractual or tariff obligation for existing generators to only sell energy to the grid. They can sell to any willing customer.[88]

Nor should the existing generator in a behind-the-meter co-location configuration with load be treated as if it is retiring for study purposes. If anything, they should be treated like capacity resources in PJM that convert to energy only.[89] The Independent Market Monitor performs a market power analysis but there is no reliability analysis. The change in status is reflected in PJM's models and studies. For an existing generator seeking to serve non-network load, PJM's Necessary

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[86] Constellation's Show Cause Answer Apr. 23, 2025 at 88 (quoting Kormos Affidavit at ¶ 6).

[87] Any argument that the Susquehanna behind-the-meter configuration had a negative effect on reliability is dispatched by the operational record once the protective relay scheme was properly installed and tested.

[88] Such a rule would also constitute an unconstitutional taking under the Fifth Amendment because, by imposing cost-prohibitive upgrades, it would effectively prevent an existing generator from selling to a willing customer and compel the generator to sell to the grid. *See Cedar Point Nursery v. Hassid*, 594 U.S. 139, 149 (2021) ("Whenever a regulation results in a physical appropriation of property, a per se taking has occurred."); *Horne v. Dep't of Agriculture*, 576 U.S. 351, 364-65 (2015) (holding that "a governmental mandate to relinquish specific, identifiable property as a 'condition' on permission to engage in commerce effects a per se taking"); *cf. Duquesne Light Co. v. Barasch*, 488 U.S. 299, 308 (1989) ("If the rate does not afford sufficient compensation, the State has taken the use of utility property without paying just compensation and so violated the Fifth and Fourteenth Amendments.").

[89] *See* PJM OATT, Att. DD § 6.6.

Studies process is the correct framework to study the reliability impact of the existing generator's modified interconnection to the grid (to accommodate the co-located load), *not* to treat the generator as if it is gone.

Replacing PJM's Necessary Study process with RMR/SSR retirement studies that would impose broad new network costs on a class of existing resources for the simple act of serving a customer through a co-located configuration chosen by the customer, which may be the only option to serve the load in a timely fashion. It would effectively preclude some customer configurations that may be more efficient and expedient and dictate a generator's perpetual service to the monopoly-controlled grid. By unjustly burdening co-location configurations, it would act in direct conflict with the principles of the ANOPR, undermine incentives to locate load as close as possible to generation, and create a powerful disincentive for any load to come online on a merchant-only basis.

Third, it is unduly discriminatory in violation of section 206 of the FPA to burden existing generation seeking to serve non-network load with extra costs and requirements through policies like heightened reliability study requirements when the Necessary Study process is sufficient to ensure reliability.[90] It is highly discriminatory to charge these costs to resources entering into such arrangements compared to resources that are actually retiring altogether which are not burdened with these costs. We are in an era—as identified in the ANOPR—"which will require *unprecedented and extraordinary quantities of electricity* and substantial investment in the Nation's interstate transmission system."[91] America needs every megawatt to serve AI infrastructure, whether in-front-of- or behind-the-meter. Undue discrimination against existing

[90] 16 U.S.C. § 824e(a).

[91] Secretary's Letter at 2 (emphasis added).

resources in the interconnection of large loads will inevitably contribute to premature retirements, discourage generation from coming online on a merchant basis, and likely will hit dispatchable and baseload units the hardest.

In the PJM Show Cause proceedings, the Commission should find that PJM's existing "Necessary Studies" process satisfies the reliability concerns animating Principle 10, and that a generator should be on the hook for the costs identified by a traditional Necessary Study. If the Commission proceeds without making this clarification, then, at a minimum, the Commission should recognize that commitments to extend the operating license of an existing resource (without which license the facility would shut down), and significant investments that expand operational capacity, should result in the existing resource being treated comparably to a new resource. Life extensions and significant uprates prevent the existing capacity from retiring and provide the same resource adequacy and affordability benefits as construction of a new resource.[92]

Since the ANOPR further prescribes that the reliability "type study" "must consider system conditions, including forecasted load growth, at least three years after the proposed suspension date" and "invite[s] comments on whether and how resource adequacy should be considered in the SSR/RMR type study,"[93] Constellation further highlights its improved load forecasting, demand side products, and reliability backstop proposals in its Joint Stakeholder Package pending in PJM

[92] The Commission has found just and reasonable ISO New England rules that treat existing resources that have undertaken specific investments like new generating capacity resources. *See ISO New England Inc.*, 119 FERC ¶ 61,045, at P 219 (2007) (approving ISO-NE Tariff, § III.13.1.1.1.2 which "enable[s] a resource that has previously been counted as a capacity resource (including one that had been deactivated or retired) to participate in a Forward Capacity Auction as a new generating capacity resource").

[93] ANOPR at ¶ 27.

and discussed below (see Principle 14).[94] These measures address the ANOPR's flagged issues without undue discrimination.

Principle 11—Basing Transmission Rates on the Quantity of Capacity and Energy Being Transmitted Across the Transmission System to the Load

The ANOPR states that "utilities serving large loads, including those at hybrid facilities, should be responsible for transmission service based on their withdrawal rights, as that value amount reflects the quantity of capacity and energy that is being transmitted across the transmission system to the load."[95] Constellation agrees that transmission rates for large loads and hybrid facilities should "reflect[] the quantity of capacity and energy that is being transmitted across the transmission system to the load," but this is not necessarily equivalent to withdrawal rights. Again, it depends on the co-location configuration.

Constellation provided detailed record evidence on appropriate rates under all potential co-location configurations in the PJM Show Cause proceedings including PJM's eight options, all depending on the level of service that the large loads and hybrid facilities desired. Zachary Ming of Energy+Environmental Economics, an expert in electricity market design who teaches a graduate course on this topic at Stanford University, summarized the three basic types of transmission service: firm standard network service, generator-contingent or behind-the-meter service, and interruptible (curtailable) service. Each of PJM's proposed options for large load configurations in those proceedings was a variation of these three basic types of service:

- firm service where the load always has rights to transmission service, the most relevant being PJM Option 2, NITS with in-front-of-the-meter co-location to the

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[94] *See* Exh. 1 (Joint Stakeholder Package).

[95] ANOPR at ¶ 28.

interstate grid (equivalent to "hybrid facilities" in the ANOPR where large load and generation share a point of interconnection to the grid);

- generator-contingent service where the load relies on co-located generation, comprised of PJM Option 4 (co-located behind-the-meter load) and Option 5 (same but with the option to purchase as-available energy); and
- interruptible service where the load can be curtailed, PJM Option 7 (NITS with co-location where load is interruptible, equivalent to ANOPR's Principle 7 on curtailable load).[96]

For PJM Option 2, or hybrid facilities as defined in the ANOPR (generation and load sharing a point of interconnection "directly to transmission facilities," or in-front-of-the-meter to Commission-jurisdictional interstate transmission facilities),[97] the "quantity of capacity and energy … being transmitted across the transmission system to the load" would be the net after subtracting injections by the generator. There already is extensive record evidence in the PJM Show Cause proceedings to demonstrate hybrid facilities' "billing should be based on the load's net use of the transmission grid, and this is true whether the load is 'large' or 'small.'"[98] A "gross load construct would apply transmission charges to the load-only portion of the co-located customer" but a "net load construct is consistent with measuring the actual energy flowing between the entire co-location configuration and the transmission system…."[99]

PJM, moreover, already has numerous behind-the-meter-generation facilities that are billed on a net basis.[100] The Commission found that BTMG netting was appropriate to "encourage

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[96] Of PJM's other options, Option 1 (firm NITS without co-location) and Option 8 (interruptible NITS with co-location where load participates as demand response) are existing Tariff services and require no Tariff changes; PJM Option 3 (firm load customer with behind-the-meter generation) also is an existing Tariff service but PJM has asserted it does not apply to large loads. Only PJM Option 6 (firm NITS with co-location plus BYOG but only for new generation) was unjust and unreasonable as proposed because it unduly discriminated against existing resources.

[97] *See* ANOPR at ¶¶ 12, 18.

[98] Constellation's April 23, 2025 Show Cause Answer at 45 (citing Ming Testimony at 55-56).

[99] Ming Testimony at 10.

[100] April 23, 2025 Answer at 42 (the existing BTMG provisions).

qualifying entities with behind the meter generation to reduce their use of the PJM transmission system."[101] In another case involving ISO New England Inc. (where netting also is standard for behind-the-meter configurations), the Commission found that netting "reasonably reflect[s] each Network Customer's usage of the transmission system and assigns the cost of providing Regional Network Service accordingly."[102] Likewise, "[e]nergy charges should be assessed to co-located generators and loads [hybrid facilities] based on the net injection to or withdrawal from the transmission system."[103]

For firm service to hybrid facilities (PJM Option 2), Mr. Ming testified that net charges were appropriate (rather than gross), but also discussed a potential "standby charge" since "there is a high likelihood that the co-located generator will be available during the coincident peak hours, meaning that the co-located load is unlikely to be withdrawing energy during coincident peak hours" and "conventional NITS charges based on net system energy withdrawals may not fully capture the costs associated with designing the transmissions system to ensure the co-located load could be served."[104] This standby charge would "reflect the fact that the transmission system must be built to a certain extent to serve the load on a firm basis."[105] Mr. Ming provided one potential methodology to calculate the standby charge, but it would be based on a "customer-designated firm load subscription level of what the customer could draw from the transmission system."[106]

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[101] *PJM Interconnection, L.L.C.*, 107 FERC ¶ 61,113, at PP 27-28 (2004) (citing *Occidental Chemical Corp. v. PJM Interconnection, L.L.C.*, 102 FERC ¶ 61,275, at P 14 (2003)) (emphasis added).

[102] *ISO New England Inc.*, 178 FERC ¶ 61,086, at P 49 (2022).

[103] Ming Testimony at 45.

[104] *Id.* at 28.

[105] *Id.* at 29

[106] *Id.*

Mr. Ming's proposal was that the load in a hybrid facility configuration would pay the *greater of* the standby charge or its net transmission rate.[107]

For generator-contingent behind-the-meter co-location configurations (PJM Options 4 and 5), the load would pay a one-time system impact charge based on the results of the PJM Necessary Study with any as-available energy purchases billed at a discounted $/MWh charge.[108] Charges for curtailable service would be based on any as-available energy purchases billed at a discounted $/MWh charge.[109] Under both generator-contingent and interruptible configurations, the load likely would have zero withdrawal rights and would not rely on "capacity and energy … being transmitted across the transmission system to the load."[110]

By contrast, Transmission Owners in PJM have sought to impose one-size-fits-all firm transmission service (as discussed above under Principle 5) and full NITS charges based on gross load no matter how much capacity and energy is being transmitted to the load. The Commission already unanimously rejected efforts by Exelon to "clarify" the PJM Tariff to change the definition of "Network Load" to require all new load to be NITS customers, finding that Exelon lacked the authority to revise PJM's Tariff.[111] No party sought rehearing. Charging all large loads a "NITS rate [network transmission rate] based on gross load (as opposed to net) no matter their actual use of the transmission system discriminates because such a construct 'is incapable of differentiating customers who impose different costs on the transmission system through co-location.'"[112]

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[107] *Id.* at 31.

[108] Constellation's April 23, 2025 Show Cause Answer at 65.

[109] *Id.*

[110] *See* ANOPR at ¶ 28.

[111] *Atl. City Elec. Co.*, 190 FERC ¶ 61,109.

[112] Constellation's April 23, 2025 Show Cause Answer at 32, citing Ming Testimony at 36.

Constellation urges the Commission to reject a one-size-fits all model for services and rates to large loads. Instead, the Commission should allow the transmission services customers want and rates that reflect the quantity of capacity and energy being transmitted to them over the grid.

Principle 12—Charging for Ancillary Services Based on Peak Demand

The ANOPR proposes that "utilities serving large loads … should be responsible for ancillary services based on peak demand, *without consideration of co-located generation*. Any co-located generating facilities will similarly be fully compensated for the provision of ancillary services."[113] Constellation does not oppose this principle.[114]

Principle 13—Establishing a Timely and Responsive Implementation Plan

The ANOPR's thirteenth principle provides that "there must be a plan to implement these proposed reforms, including the treatment of large load interconnections that are already being studied for interconnection."[115] The Commission should of course develop such a plan, but as detailed in Section II below, the Commission should begin by taking immediate action in the long-pending PJM Show Cause (and related) proceedings by ruling on the merits and expediting implementation in PJM. Alternatively, if the Commission believes that the ANOPR adds new issues that were not already sufficiently addressed in record in the Show Cause proceedings or through additional comments submitted in response to this ANOPR, the Commission should issue a targeted order in the PJM Show Cause proceedings that eliminates ambiguity and delay in three key areas related to jurisdiction, rates, and timeliness so that more large load projects can move forward. With respect to the "the treatment of large load interconnections that are already being

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[113] ANOPR at ¶ 29 (emphasis added).

[114] Separate ancillary service rate arrangements are justified for behind-the-meter large loads that are not connected to the grid. *See* Constellation's April 23, 2025 Show Cause Answer.

[115] ANOPR at ¶ 30.

studied for interconnection" in PJM, the Commission should immediately put a halt to any PJM Transmission Owner's refusal to proceed with the interconnection studies of co-location configurations (Constellation proposes specific measures in Section II).

The Commission should reject any argument that it must wait to rule in PJM until a final rule on the ANOPR is issued. While the Commission's ruling in PJM clearly will inform the Commission's deliberations on other large load matters including the ANOPR, the Commission can and should undertake those deliberations in parallel to acting on the pending cases under section 206 of the FPA that have long been pending in PJM given the importance of this region and the national interests at stake. PJM cannot be allowed to have unclear rules and stalemate any longer. To the extent that the Commission's later action in response to the ANOPR requires further work in PJM, the Commission can direct PJM to complete that work as part of the compliance process in that proceeding.

As for a transition plan for "the treatment of large load interconnections that are already being studied for interconnection,"[116] the Commission should grandfather interconnection studies that already are substantially completed by the time the Commission accepts final compliance filings in response to the Commission's final rules.[117] It would be inefficient and cause unnecessary cost and delay to require these projects where studies have long been pending or are nearly completed to start over in the study process.

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[116] ANOPR at ¶ 30.

[117] The Secretary called for final rules by April 30, 2026. RTOs and Transmission Owners presumably will have to submit compliance filings in response to the final rules.

__Principle 14—Meeting Reliability Standards__

The ANOPR states that "utilities serving large loads must meet all applicable NERC reliability standards and OATT provisions."[118] Constellation agrees. To the extent the Commission has concerns that large loads may jeopardize reliability, Constellation proposes measures that can remedy reliability concerns without delaying large load interconnections.

It must be remembered that one of the primary benefits of the co-location of large loads and generators is that it improves reliability. Co-location that "avoids the use of the grid"—such as a large load taking its power directly at a shared point of interconnection with a generator as hybrid facilities—minimizes "grid operational risks (because fewer grid infrastructure facilities and potential failure points are implicated in serving the load)," while also minimizing "transmission infrastructure costs, potential supply chain issues and delays on long-lead transmission infrastructure equipment ... [and] line losses."[119] The same logic to applies to both *new* and *existing* generation as the same benefits are realized by co-locating with an existing generator. One of the primary attributes of co-location configurations (all of them) is that they take advantage of existing infrastructure and therefore minimize costs.

The ANOPR's fourteenth principle on reliability further states that "[u]tilities and we must be prepared to revise large load interconnection procedures and agreements, as necessary,"[120] presumably to incorporate reliability protections. This is exactly what Susquehanna Nuclear attempted to do over a year ago with its modified interconnection agreement to incorporate a co-

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[118] ANOPR at ¶ 31.

[119] Affidavit of Michael Kormos at ¶ 4 (Exh. 5 to Constellation's April 23, 2025 Show Cause Answer); *see also* ANOPR at ¶ 20 (highlighting that "siting a large load near or at the same point of interconnection as a new generating facility could reduce the network upgrades needed to interconnect only the load or only the generating facility").

[120] ANOPR at ¶ 31.

located large load.[121] Other utilities in PJM have since refused to process interconnection modification requests for co-location projects. As set forth below, the Commission should immediately require PJM Transmission Owners to cooperate in current study processes designed to ensure reliability, but should also confirm timely utility cooperation is required in any final rule.

While ensuring the reliability of the grid is essential, large loads also *must be served*, and as quickly as possible, consistent with system reliability. Utility regulation is founded on the principle that utilities authorized to operate monopolies must submit to regulation, including price regulation and the obligation to extend service to customers within their service territory and to provide continued service once commenced.[122] The latter is commonly referred to as the "duty to serve." Nearly every jurisdiction in the United States has promulgated regulations establishing a duty to serve.[123] Although each state applies its own flavor, the general requirement is the same—an electric utility must serve customers seeking service and cannot unduly discriminate or delay such service.[124] Illinois's statute, for example, provides that the duty to serve is "without discrimination and without delay."[125] Delays of five, seven, or ten years—which are commonplace when trying to connect a large load to the grid—are untenable and objectively cannot be considered

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[121] *PJM Interconnection, L.L.C.*, 191 FERC ¶ 61,025, at P 30 (2025). The Commission rejected amendments to a Commission-jurisdictional interconnection agreement to accommodate a hybrid facility configuration. Nearly all these amendments dealt with reliability issues.

[122] Jim Rossi, *Universal Service in Competitive Retail Electric Power Markets: Whither the Duty to Serve?*, 21 ENERGY L. JOURNAL 27 (2000).

[123] *See* Heather Payne, *Unservice: Reconceptualizing the Utility Duty to Serve in Light of Climate Change*, 56 UNIV. OF RICH. L. REV. 603, 610-12 (2022) (describing the duty to serve in various states).

[124] *See*, *e.g.*, 220 Ill. Comp. Stat. 5/8-101 (Illinois statute stating that a "public utility shall, upon reasonable notice, furnish to all persons who may apply therefor and be reasonably entitled thereto, suitable facilities and service, without discrimination and without delay"); N.J. STAT. ANN. § 48:3-3(a) ("No public utility shall provide or maintain any service that is unsafe, improper or inadequate, or withhold or refuse any service which reasonably can be demanded or furnished...."); VA. CODE ANN. § 56-234(A) ("It shall be the duty of every public utility to furnish reasonably adequate service and facilities at reasonable and just rates to any person, firm or corporation along its lines desiring same.").

[125] 220 Ill. Comp. Stat. 5/8-101.

service "without delay." While "the demand for interconnection by data centers driven by the needs of artificial intelligence and other technologies is a recent phenomenon," the obligation to serve load without delay is as old as utility regulation.[126]

The FPA also prohibits undue discrimination, with specific provisions against discrimination during shortage conditions such as those identified in the ANOPR.[127] It is unlawful, for example, to decide not to serve large loads to ensure service to others, or to favor new generation over existing generation in taking measures to remedy the shortage.[128]

Another fact to keep in mind is that PJM and other regions already have extensive market design mechanisms, planning requirements, and other carefully developed and scrutinized processes to ensure both reliability and resource adequacy (which refers to sufficient capacity and is not the same as reliability). This includes existing interconnection study processes that already effectively analyze the reliability effects of interconnections.

If the Commission is concerned that these reliability tools are insufficient, the solution cannot be to prevent service to large loads or to delay service for years. The Commission could instead improve reliability by adopting the Joint Stakeholder Proposal that Constellation along with Amazon, Calpine, Google, Microsoft and Talen proposed, including: (1) load forecast and other modeling improvements, (2) limited demand response products that align with the curtailments or operation of backup generation that the large loads are willing and able to provide, and (3) reliability backstop mechanism enhancements if a forward capacity auction clears

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[126] Answer and Motion to Dismiss of PJM Interconnection, L.L.C., Docket No. EL25-20-000, at 5 (filed Dec. 12, 2024).

[127] *See* 16 U.S.C. § 824a(g)(3)(B).

[128] *Id.* (the "manner" of addressing "any anticipated shortage of electric energy or capacity" to "insure continuity of service to customers," must ensure that all persons served by a public utility are treated "*without undue prejudice or disadvantage*") (emphasis added).

insufficient capacity to meet the forecast to ensure there is sufficient available generation if the market mechanism fails to do so.[129]

Load Forecasting and Modeling Improvements

One suite of enhancements goes to the persistent problem of load forecasting. Our proposal is to:

- Include only large new loads backed by meaningful, verifiable commitments such as executed energy-service agreements, posted credit or collateral, significant infrastructure investment, or long-term supply commitments with new or existing resources.
- Use realistic ramp-rate trajectories, utilization factors, and safeguards to avoid double-counting across overlapping proposals or procurement paths.
- Require aggregate large-load forecasts to account for project-realization indicators such as historical success rates, development progress, supply-chain constraints (e.g., chip availability), independent expert assessments, and structured information exchange among large-load customers, PJM, and Transmission Owners.

New Demand Side Products

One way of increasing the capacity available to the market is to allow for limited demand response products more aligned with the capability of data centers and their backup generation. For example, in the stakeholder process, Constellation supported new means of obtaining demand-side resource adequacy services from large loads (including data centers) that could be offered into the capacity market, thereby allowing large loads to tap into their already-available operational flexibility to provide resource adequacy services in response to system stress. Large loads could voluntarily submit offers into PJM's capacity auction to serve as limited demand response resources under one of two new limited demand response categories: (i) a maximum of 24 hours/year of curtailment with no single interruption greater than 6 hours and (ii) a maximum of

[129] *See* Exh. 1 (Joint Stakeholder Package).

100 hours/year of curtailment with no single interruption greater than 10 hours. PJM would develop a new Effective Load Carrying Capability (ELCC) for each category allowing large load capacity market offers and, if cleared, performance when dispatched by PJM (consistent with the limitations associated with their demand response category).

Reliability Backstop Enhancements

The Joint Stakeholder Package also proposed enhancements to PJM's existing Reliability Backstop Mechanism to provide further incentives for new resources to enter the market when the supply and demand balance is particularly tight. Under this enhancement, eligible resources, including new/reactivated generation, existing generation with an offer cap above the highest point on the Variable Resource Requirement ("VRR"), and traditional demand response, would have the option to offer a minimum term supply commitment, priced at the top of the VRR for a multi-year period. Resources cleared by PJM in this fashion would receive a capacity commitment for the term of years and price offered. The features of this enhancement were carefully structured to respond to only legitimate resource adequacy deficits but to minimize overreliance on potentially lower quality and/or out-of-market services which could have a long-term detrimental impact on the market and its ability to signal investment when needed. The following elements were included in this proposal to strike this balance:

- To ensure the mechanism is only available in years with a meaningful potential shortfall, the reliability backstop alternative would only be triggered in years when RPM would otherwise clear below 98% of the reliability requirement.

- Bidders may offer up to 7 years, but shorter-term offers will clear first to minimize the duration of any market intrusion. PJM would stop clearing reliability backstop resources once a cleared volume clears above 98% of the reliability requirement.

- To ensure "phantom" capacity is not offered, eligible resources must be reasonably likely to perform in the delivery years cleared.

- To minimize the impact on clearing prices for the rest of the market, in subsequent years, cleared resources with multi-year commitments will be entered into the RPM potentially based on the technology-specific default gross avoidable cost minus the unit-specific projected Energy and Ancillary Services ("EAS") offset.

- To ensure any winning offers do not receive a windfall, if the market clearing price in subsequent years is different than their original offer price of cleared multi-year resources, the resource will receive the multi-year price agreed – no higher/lower.

- To ensure the mechanism does not outlast its need, it will be in place for the four BRAs corresponding to the current Quadrennial Review (2028/29 through 2031/32) and will automatically sunset thereafter.

The Commission should adopt measures like these in a final rule and in an order in the PJM Show Cause proceedings. The solution to reliability concerns cannot be to allow service to large loads to be delayed for years.

II. The Commission Should Find the Existing PJM Tariff Unjust and Unreasonable and Provide Immediate Relief

A. The Commission Should Immediately Issue a Full Order on the Merits in the PJM Show Cause Proceedings

Consistent with the ANOPR and based on the comprehensive record it has assembled in three separate proceedings, comprised of a fully-briefed complaint proceeding, a national inquiry into co-location issues, and another fully-briefed Show Cause proceeding (with 38 itemized briefing questions), the Commission should issue a full order in the PJM Show Cause proceedings confirming its preliminary finding that the PJM Tariff is unjust and unreasonable because it does not contain "provisions addressing with sufficient clarity or consistency the rates, terms, and conditions of service that apply to co-location arrangements"[130] and "[t]he absence of such provisions may leave entities unable to determine what steps they can or must take to effectuate

[130] *PJM Show Cause Order*, 190 FERC ¶ 61,115 at P 74.

co-location arrangements of various configurations."[131] That reasoning remains just as true today, and the PJM Tariff's lack of clarity continues to impede timely large load development.

The comprehensive record in the PJM Show Cause proceedings also supports a Commission ruling on a full replacement rate. That replacement rate should modify the existing PJM Tariff to provide for a variety of co-location options to provide strong incentives for large new load to locate in places and on terms that maximize the efficiency of the existing transmission grid and allow for timely service without jeopardizing reliability or unjustly shifting costs. PJM has identified eight potential options for how large, new load might connect to the interstate grid:

- PJM Option 1: Firm NITS without co-location (*i.e.*, standard service);
- PJM Option 2: Firm NITS with in-front-of-the-meter co-location (equivalent to "hybrid facilities" in the ANOPR where large load and generation share a point of interconnection to the grid);
- PJM Option 3: Firm NITS with the generation behind-the-meter of the load with transmission charges based on net usage of the grid (existing "BTMG" under the PJM Tariff, which PJM claims is limited to small loads);[132]
- PJM Option 4: Co-located behind-the-meter load with "generator-contingent" protections;
- PJM Option 5: Co-located behind-the-meter load with "generator-contingent" protections plus as-available energy purchase option;
- PJM Option 6: Firm NITS with co-location plus expedited generation interconnection for BYOG;
- PJM Option 7: NITS with co-location where load is interruptible (equivalent to ANOPR's Principle 7 on curtailable load);[133] and

[131] *Id.*

[132] *Contra* Constellation's April 23, 2025 Show Cause Answer at 44 ("there is no size limitation for BTMG in the PJM Tariff, and it has been implemented for large loads").

[133] The Commission should clarify that provisional *interruptible* services (PJM Option 7 with Constellation's proposed rate treatment) end once the network upgrades are built. *See* Constellation's May 8, 2025 Answer at 14.

- PJM Option 8: NITS with co-location where load participates as demand response. (Constellation proposes additional demand response measures in the Joint Stakeholder Package).[134]

Constellation explained how PJM options 2-5 and 7 were viable replacement rates, with some clarifications and modifications, including to the appropriate rate treatment, and the Commission should order PJM to implement them in a compliance filing within 60 days.[135]

In particular, the Commission should:

- Adopt a replacement rate that allows for PJM's proposed co-location configuration options, including the equivalent of the ANOPR's "hybrid facilities" (as defined in the ANOPR, equivalent to PJM Option 2), and the ANOPR's curtailable load (equivalent to PJM Option 7), as well as behind-the-meter configurations (PJM Options 4 and 5), as clarified and modified by Constellation, with rate treatments that recognize the different uses of the transmission system of these different configurations,[136] and direct PJM to implement the options in a compliance filing within 60 days.[137]

- Reject Bring-Your-Own-Generation (PJM Option 6) as unduly discriminatory and preferential because only new generation would qualify.[138]

- Find that the Commission does not have jurisdiction over behind-the-meter sales between a co-located generator and load, regardless of whether the load is "synchronized" to the grid.[139]

- Order Transmission Owners to provide timely participation in relevant studies requested by PJM and in the interconnection agreement amendment process.[140]

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[134] *See* Exh. 1; *see also supra* at Principle 14 (discussing same).

[135] Constellation's April 23, 2025 Show Cause Answer at 6; *see also id.* at 50; *see also* Constellation's May 8, 2025 Preliminary Finding Answer at 14-15.

[136] Constellation's April 23, 2025 Show Cause Answer at 37-38, 65, 108. Note that PJM Option 1 (firm NITS without co-location), Option 3 (co-location with behind-the-meter-generation), and Option 8 (firm NITS with demand response) all are already in the PJM Tariff. The Commission should clarify that provisional interruptible services (PJM Option 7 with Constellation's proposed rate treatment) ends once the network upgrades are built. See Constellation's May 8, 2025 Preliminary Finding Answer at 14.

[137] Constellation's April 23, 2025 Show Cause Answer at 6, 52-55; *see also* Constellation's May 8, 2025 Preliminary Finding Answer at 14-15.

[138] Constellation's April 23, 2025 Show Cause Answer at 52-55.

[139] *Id.* at 71-73; *see also* Herling Affidavit at ¶ 33 (explaining synchronization does not imply transmission usage).

[140] Constellation's April 23, 2025 Show Cause Answer at 89.

- Direct Transmission Owners to process load interconnection requests on the same timelines regardless of configuration to avoid the exact type of discrimination that led to Order No. 2003 for generator interconnections.

Some will argue that facilitating large load interconnections is a threat to resource adequacy and that a full order on the merits in PJM is premature. Constellation recognizes the resource adequacy concerns associated with significant load growth. But resource adequacy concerns are not a reason to leave interconnection rules undefined, effectively ceding regulatory control to Transmission Owner discretion. Nor are they an excuse to discriminate against existing generation in favor of new resources.

Nevertheless, to the extent that the Commission determines reliability enhancements are required, the Commission should not delay a ruling in PJM but should proceed on its own expedited track. For example, Commission could order PJM to submit compliance filings on the Joint Stakeholder Package's three reliability-enhancing measures that Constellation has proposed (jointly with Amazon, Calpine, Google, Microsoft and Talen).[141]

The Commission should also direct PJM to propose a standard large load interconnection process of 3-6 months for interconnections to transmission facilities,[142] consistent with the ANOPR's core expediency concerns. This basic timeline is imperative to get projects moving through the interconnection process.

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[141] *See* Exh. 1 (Joint Stakeholder Package); *see also supra* at Principle 14 (discussing same).

[142] *See, e.g.*, Pa. Pub. Utils. Comm'n, Interconnection and Tariffs for Large Load Customers, Docket No. M-2025-3054271, Tentative Order at 25 (Nov. 6, 2025), https://www.puc.pa.gov/pcdocs/1901687.pdf (proposing a six-month maximum timeline for completion of interconnection studies for large load customers).

B. Alternatively, the Commission Should Immediately Issue a Targeted Order in the PJM Show Cause Proceedings to Jumpstart Large Load Development

In the event the Commission determines that it needs more time to issue a comprehensive order on the merits in the PJM Show Cause proceedings at this time, it should still at a minimum issue an order confirming its preliminary finding that the PJM Tariff is unjust and unreasonable and providing targeted rulings consistent with the ANOPR's principles on three core issues to ensure that development of large loads proceeds without further delay: (1) the Commission's jurisdiction includes the interconnection to the grid by hybrid facilities; (2) NITS charges for hybrid facilities should be based on the load's net usage of the transmission system; and (3) PJM and Transmission Owners must immediately act on large load studies and process large load interconnection requests, including those that have long been pending, to ensure timely interconnection of large loads to the grid.

1. The Commission Should Affirm Jurisdiction Over Hybrid Facilities

As discussed above, the ANOPR correctly notes the Commission's jurisdiction over large load interconnections to transmission facilities.[143] The jurisdictional boundaries relating to large load interconnections have been extensively analyzed by the Commission in the PJM Show Cause Order, briefed by the scores of parties in PJM Show Cause proceedings in response to the Commission briefing questions, and now further analyzed by the Secretary in the ANOPR.[144] Yet the lack of a Commission determination is enabling utilities to delay the processing of large load

[143] ANOPR at ¶ 18; *see also* Secretary's Letter at 1 ("It is my view that the interconnection of large loads directly to the *interstate transmission system* to access the transmission system and the electricity transmitted over it falls squarely within the Commission's jurisdiction.") (emphasis added).

[144] *See* PJM Show Cause Order, 190 FERC ¶ 61,115 at P 88 (Briefing Question a.i. on "jurisdictional principles"); *see also* Constellation's April 23, 2025 Show Cause Answer at 68-74 (answering).

interconnection requests and use their control over the process to force load to accede to distribution service or configurations that make netting impossible.

Accordingly, the Commission should issue an order in PJM that at a minimum finds that the interconnection of hybrid facilities—where a large load and a generator share a point of interconnection—is FERC-jurisdictional. The Commission could make the same finding with respect to all large load interconnections connecting to the interstate grid and taking transmission service, but to the extent the Commission determines it needs further briefing on some large load interconnections, it should still address the interconnection of hybrid facilities—facilities over which the Commission already has asserted jurisdiction with respect to the generator connected to them.

2. *The Commission Should Rule that Hybrid Facilities Should Pay Net NITS*

The Commission should rule that the appropriate transmission rate for hybrid facilities sharing an interconnection to the transmission grid is based on the load's net usage of the transmission system, "reflect[ing] the quantity of capacity and energy that is being transmitted across the transmission system to the load."[145]

Constellation has long advocated that transmission service based on "net" usage of the transmission system is the correct rate for an in-front-of-the-meter co-location configuration where the load and generator share the same interstate point of interconnection (what the ANOPR calls "hybrid facilities" and equivalent to PJM Option 2 in the PJM Show Cause proceedings). As set forth in detail below, Constellation has also explained why it may also be appropriate to charge the *greater* of net withdrawals or a standby charge for hybrid facilities sharing an interconnection

[145] ANOPR at ¶ 28. As set forth in our discussion of Principle 11, the ANOPR appears to equate withdrawal rights with the quantity of capacity and energy being transmitted to a large load or hybrid facilities. They are not always equivalent.

to the interstate grid. This issue of netting has been a long-standing point of disagreement in the ongoing complaint, Show Cause and other proceedings—including one Commission-rejected attempt by Transmission Owners to amend the PJM Tariff to require gross charges[146]—that can be quickly resolved through the PJM Show Cause proceedings. For purposes of an immediate order in PJM, the Commission should affirm NITS charges for hybrid facilities should be based on net transmission usage, not gross.

3. The Commission Should Require PJM and Transmission Owners to Immediately Act on Large Load Studies and Interconnections

The ANOPR's central objective is "to ensure the timely and orderly interconnection of large loads."[147] Accordingly, the Commission should order Transmission Owners to timely process large load requests. Currently, they are not all doing so. Large load projects that are moving forward are either the transmission utilities' own projects, or projects where the customer agrees to connect in exactly the configuration that the utilities—in their unilateral discretion—require. This is untenable and is blocking progress on critical infrastructure development needed to facilitate AI. It also is contrary to the core objectives of Order No. 888 to stop Transmission Owners from granting faster service to their own or to favored projects.[148]

In the PJM Show Cause Order it issued nine months ago, the Commission preliminarily found that the "lack of applicable [PJM] Tariff provisions raises the potential for undue discrimination or preferential treatment," citing in support record evidence of Exelon's refusal to

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[146] *Atl. City Elec. Co.*, 190 FERC ¶ 61,109.

[147] ANOPR at ¶ 1.

[148] *Promoting Wholesale Competition Through Open Access Non-discriminatory Transmission Services by Public Utilities and Recovery of Stranded Costs by Public Utilities and Transmitting Utilities*, Order No. 888, FERC Stats. & Regs. P 31,036 (1996), *order on reh'g*, Order No. 888-A, FERC Stats. & Regs. P 31,048 (1997), *order on reh'g*, Order No. 888-B, 81 FERC ¶ 61,248 (1997), *order on reh'g*, Order No. 888-C, 82 FERC ¶ 61,046 (1998).

process Constellation's co-location projects because—according to Exelon—co-location of load and generation is "not allowed under the Tariff without arranging for wholesale transmission or retail distribution service."[149] Nothing relevant to the Commission's prior preliminary finding has changed since then—despite the fact that the PJM Tariff already states that PJM "is responsible for the preparation of all studies required by the Tariff … [and] Transmission Owner(s) shall supply such information and data reasonably required."[150]

Accordingly, the Commission should immediately order the following:

- PJM must establish a timeline – with milestones – for the completion of load interconnection studies for hybrid in-front-of-the-meter configurations (those where load is connecting at the same point on the transmission system as the existing or new generation), with an absolute study completion deadline, including Transmission Owner input, of six months.

- PJM must require that Transmission Owners comply, cooperate in, and provide timely feedback on the Necessary Studies and other processes to ensure completion within PJM's timeline for large load interconnection requests that are behind-the-meter arrangements.

- PJM must compile and maintain a list of all pending large load interconnection requests (over a certain size threshold and with defined meaningful/verifiable commitments) and require progress updates from the applicable Transmission Owner every 30 days.

- PJM must file a quarterly status report with the Commission on all pending large load interconnection requests and Necessary Studies. In the same docket, PJM must submit a monthly information filing to the Commission in the event that any Transmission Owner misses a milestone or fails to make meaningful progress on any pending large load interconnection request in the previous 30 days. The Commission should require Transmission Owners to explain the lack of progress within five days.

- No Transmission Owner can expedite service or study processing for projects taking the Transmission Owner's preferred type of service or configuration, or refuse to process a complete large load interconnection request on the ground that

[149] PJM Show Cause Order, 190 FERC ¶ 61,115 at P 75.

[150] PJM OATT Part VII, § 322.

the large load has not "arranged for wholesale transmission or retail distribution service."

- Transmission Owners will be subject to referral to the Commission's Office of Enforcement for any failure to diligently process large load interconnection requests or for conditioning the processing of such requests on the applicant's acceding to the Transmission Owner's favored configuration or unnecessary services, so that the Commission can investigate any Tariff or statutory violations (including undue discrimination or preference, manipulation, or false communications).
- Clarify that Transmission Owners must complete any large load interconnection studies and Necessary Studies already in progress, and any studies that are substantially completed when the Commission accepts final compliance filings on large load interconnection rules need not restart the study process.

These requirements would immediately get numerous data center projects back on track.

While the Commission has the record to rule on the merits of every issue in the PJM Show Cause proceeding, and should do so, to the extent the Commission decides otherwise, it should at least issue rulings on these three issues. Doing so would be a critical first step to implementing the ANOPR in PJM.[151]

CONCLUSION

WHEREFORE, Constellation requests that the Commission propose and implement a final large load interconnection rule consistent with the Secretary of Energy's fourteen principles, as qualified herein, by April 30, 2026. Constellation also requests an immediate comprehensive order on the merits in the PJM Show Cause proceedings as informed by the guidance in the ANOPR. To the extent that the Commission is not prepared to issue a full PJM order on the merits on the well-

[151] The Commission could also require PJM to submit a compliance filing on the Joint Stakeholder Package of three reliability-enhancing reforms. *See* Exh. 1 (Joint Stakeholder Package). The Commission should also direct PJM to propose a standard large load interconnection process of 3-6 months for interconnections to transmission facilities.

developed record in those proceedings, Constellation requests an order in PJM on (1) jurisdiction, (2) rates, and (3) timeliness.

Respectfully submitted,

<u>/s/ Paul F. Wight</u>

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November 21, 2025

Exhibit 1

Joint Stakeholder Package
PJM CIFP – Large Load Additions

Joint Stakeholder Package
PJM CIFP - Large Load Additions

Amazon, Calpine, Constellation, Google, Microsoft, Talen

November 7, 2025

Package

- Large Load Forecasting Improvements
- Demand Side
- Reliability Backstop Alternative

Large Load Forecasting Improvements

- Only include large new load that has a meaningful/verifiable commitment. Examples of which include:
  - Energy Service Agreement
  - Credit/collateral support
  - Financially significant infrastructure investment
  - Long-term supply commitment with new or existing resources
- Improve large load modeling assumptions including, but not limited to, ramp rate assumptions, utilization rates, and reduction of double counting
- Implement a Large Load Forecast "Reality Check": Consider other relevant factors that may predict whether all proposed large load is likely to come to fruition. Examples include, but are not limited to:
  - Historic large load success rate and development progress reviews along with comparison to supplemental projects
  - Supply chain and other limitations such as chip (un)availability
  - Other business expert reviews (McKinsey, S&P, BNEF, LEI Study, LBNL, etc.)
  - Conversations/information exchange among large load customers, PJM, and TOs

New Demand Side Products

- Voluntary large load limited demand response
  - Allow large load to clear as limited demand response. Discounted Effective Load Carrying Capability (ELCC) to reflect the reduced availability. Two, new limited products would be defined as:
    - 24 hours/year with no single interruption greater than 6 hours
    - 100 hours/year with no single interruption greater than 10 hours
- Back-up generator operations
  - Provide relief during system emergencies using back-up generators. Agree to operate within criteria acceptable to the owners (i.e., consistent with their commitment to the environment, noise reduction, etc.). Establish new PJM emergency procedure step, 9A, for deployment immediately prior to manual load dump. Discounted ELCC to reflect the reduced availability.

Reliability Backstop Alternative

- The following reliability backstop alternative would only be utilized if triggered.
  - Trigger: Only used in years when RPM clears below 98% of the reliability requirement.
- Avoid overreliance on potentially lower quality and/or out-of-market services which could have a long-term impact on the market and its ability to signal investment when needed. Every attempt made to minimize such impact although it can’t be eliminated.

Reliability Backstop Alternative (cont.)

- Eligible resources can (at their option) offer a minimum term supply commitment at the top of the Variable Resource Requirement (VRR) curve.
- Bidders may offer up to **7** years but shorter-term offers will clear first.
- Eligible resources include: new/reactivated generation, existing generation with an offer cap above top of the VRR curve, traditional demand response.
- Eligible resources must be reasonably likely to perform in the delivery years cleared.
- In subsequent years, cleared resources with multi-year commitments will be entered into the RPM potentially based on the technology-specific default gross avoidable cost minus the unit-specific projected Energy and Ancillary Services (EAS) offset.
- If the market clearing price in subsequent years is different than their original offer price of cleared multi-year resources, the resource will receive the multi-year price agreed – no higher/lower.
- Mechanism will be in place for the four BRAs corresponding to the current Quadrennial Review (2028/29 through 2031/32) and will automatically sunset
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Summary

Proposed Solution Options

- Improve load forecast.
- Utilize new demand side resource products.
- In the event the system remains short, utilize reliability backstop alternative.

Other Future Potential Considerations

- Extension of cap and floor, with price adjustments
- Improvement of energy and ancillary service markets

Feedback

Appendix

PJM Emergency Procedure Steps

- EEA1 = PJM Maximum Generation Emergency Alert
- Step 1: Pre-Emergency Load Management Reduction Action (30, 60 or 120-minute)
- Step 2: Emergency Load Management Reduction Action (30, 60 or 120-minute) (EEA2)
- Step 3 (real-time): Primary Reserve Warning
- Step 4 A (Real-time): Maximum Generation Emergency Action (M18 – Max Emergency Generation Action)
- Step 4 B (Real-time): Emergency Voluntary Energy Only Demand Response Reduction Action
- Step 5 (Real-time): Voltage Reduction Warning & Reduction of Non-Critical Plant Load (EEA2)
- Step 6 (Real-time): Curtailment of Non-Essential Building Load
- Step 7 (Real-time): Deploy All Resources Action
- Step 8 (Real-time): Manual Load Dump Warning
- Step 9 (Real-time): Voltage Reduction Action (EEA2)
- Step 10 (Real-time): Manual Load Dump Action (EEA3)
- Load Shed Directive